

Scientific Learning

* Exercise:

A. Choose the best alternative.

1. Which one of the following involves in practical work?

(i) ~~obers~~ observation

(ii) test

(iii) hypothesis

(iv) ☒ all of them

2. Which of the following practical activity cannot be done in laboratory?

(i) preparation of O_2

(ii) formation of crystal

(iii) ☒ observation of star(iv) preparation of H_2

3. Which of the following practical activity can be done in laboratory?

(i) making a magnet

(ii) reflection of light

(iii) observation of cell

(iv) ☒ all of them

4. Research report includes:

(i) ☒ aim

(ii) objective

(iii) ☒ hypothesis

(iv) all of these

5. What should be done while performing a chemical reaction that produces fumes?

(a) ☒ hands should be kept away (b) it should be carried out on a table.

(c) it should be kept on a stand.

(d) it should be carried in hand.

6. You have been injured in the laboratory (cut, burn etc). First you should

(a) visit the school nurse after class

(b) see a doctor after school.

(c) tell the science instructor at once

(d) ☒ apply first aid yourself

7. When you finish working with chemicals, biological specimens and other lab substances always?

- a) treat your hands with skin lotion
- ☒ b) Wash your hands thoroughly with soap and water
- c) Wipe your hands on a towel
- d) Wipe your hands on your clothes

8. Which of the following is the first step in starting the research process?

- a) survey of related literature
- ☒ b) identification of the problem
- c) searching sources of information to locate the problem.
- d) searching for solutions to the problem

9. Research is

- ☒ a) Searching again and again
- b) finding a solution to any problem
- c) working in a scientific way to search for the truth of any problem
- d) None of the above

10) A survey can collect information using which of the following techniques?

- a) telephone
- b) face-to-face interviews
- c) mail
- ☒ d) all of the above

B. Very short answer type questions.

1. What is scientific learning?

→ Scientific learning is the systematic process of gaining knowledge through observation, experimentation, and reasoning.

2. What is observation?

→ Observation is the careful study of an object, event, or phenomenon using sense organs.

3. Write two ways of observation

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- Observation by sense organs.
 - Observation by scientific tools

4. What is hypothesis?

→ A hypothesis is a tentative answer or assumption made to solve a problem.

5. What is report?

→ A report is a written record of the findings and conclusion and report writing.

6. What are the steps of scientific learning?

-
- Observation
 - Ask a question
 - hypothesis
 - experiment
 - analysis
 - conclusion
 - report writing

7. Define scientific research.

→ Scientific research is a systematic investigation carried out to discover new facts or verify.

8. Mention the steps of scientific research.

→ Observation, problem identification, hypothesis, experiment, data collection, analysis, conclusion, and report writing.

9. Give an example of scientific research.

→ studying the effect of fertilizer on plant growth.

10. ~~What~~ What is the goal of scientific research?

→ The goal of scientific research is to discover new ~~knowlt~~ knowledge and solve problems scientifically.

C. Short answer type questions

1. Write any two importance of practical work.

→ - It helps to understand scientific concepts clearly.

• It develops scientific skills and interest.

2. Write any two general laboratory safety rules.

→ • Wear a lab coat and goggles.

• Do not touch chemicals without permission of guard.

3. Write any two advantages of report writing.

→ • It keeps of permanent record of findings.

• It helps others understand the work done.

4. Mention any four steps that help in test of experiment.

→ (i) Observation

(ii) Hypothesis

(iii) Experimentation

(iv) Conclusion

5. observation is the ~~best~~ first step in scientific learning.
Give reason

→ observation helps identify problems and gather information for further study.

6. A report is prepared after the experiment and survey are done. Give reason.

→ A report is prepared after collecting and analyzing all data and results.

7. Why create surveys using survey tools?

→ survey tools help collect accurate and organized information from many people.

8. Write the difference between scientific research and survey.

scientific research	survey
(i) Conducted to discover new facts.	(i) Conducted to collect information
(ii) User experiment	(ii) User questionnaires and interviews.

9. Write the difference between observation and scientific experiment (Test).

Observation	scientific Experiment
(i) Information is collected by watching.	(i) Information is collected by testing.
(ii) No special control is required.	(ii) Controlled condition are required.

10. What are the steps taken to prepare scientific research?

→ observation, problem identification, hypothesis experiment, data collection, analysis, conclusion and report writing.

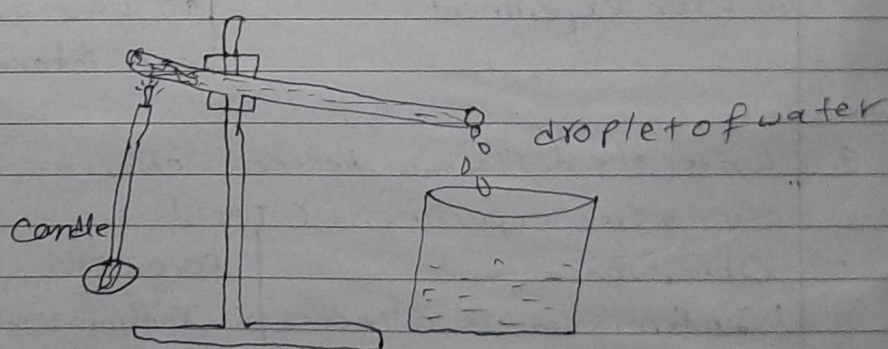
D. Long answer type questions.

1. "Scientific learning is necessary for our life?"
Justify.

→ Scientific learning helps to understand things correctly and solve problems logically. It improves our knowledge, removes superstitions, and makes life easier through inventions and discoveries. Therefore, scientific learning is necessary for our life.

2. How can you prepare crystal of copper sulphate? Explain with necessary diagram.

→ Crystal of copper sulphate are prepared through a process called crystallization. This involves dissolving copper sulphate powder in hot, acidic water until saturated, then cooling the solution slowly to allow pure, solid crystal to form.



3) ~~Q~~ How can we separate salt from salt solution? Explain with necessary diagram.

→ Salt solution can be separated from salt solution by the evaporation method.

* Procedure:

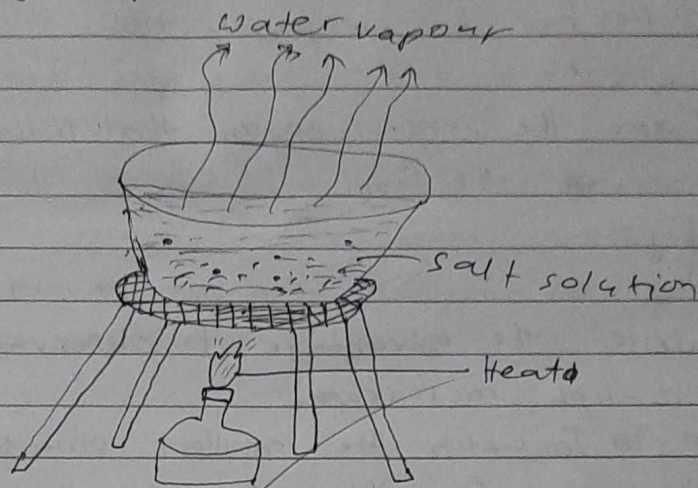
1) Take the salt solution in an evaporating dish.

2) Heat the solution gently.

3) Water evaporates into water vapour.

4. salt remains in the dish

Necessary Diagram:



4) How does scientific report help in the scientific learning process? Explain with example.

→ Scientific report help in scientific learning process by reading recording and communicating the aim, procedure, observation and result of an experiment. They helps us to understand, verify and share scientific finding.

For example a report of salt solution experiment helps to other for understand and verify the findings.

5) Nilima is a student of class 8, is observing structure of fungi through a microscope in a science laboratory but she not in proper attire and does not follow the laboratory safety rules. Study the diagram and answer the following questions.

a) Find two mistakes done by the Nilima in the laboratory on observing the structure of Fungi.

→ iv She is ~~to~~ not wearing a laboratory coat, gloves and goggles
v Her hair is not properly tied.

b) Name the sense organ that Nilima uses to observe the fungi.

→ Eye

c) Write the advantage of observation of fungi through microscope.

→ ~~to~~ To watch the smallest structures of fungi in large for study properly.

d) Name the tools that help in observation.

→ Microscope is the tools that help in observation.

e) Mention other sense organs that help in observation.

→ Skin.

6. Lab demonstrator in a science laboratory gives some common substances used in daily life to the students with some chemical tools and asked them to identify the given substance as an acid, base and salt. By the observation, some student identified them but some other students oppose their identification and go to further identification by scientific test to identify them. Study the above paragraph and answer the following questions.

a) Mention some common etc chemicals used in your daily life as an acid, base and salt.

→ Acid :- vinegar, lemon juice
Base :- soap, tooth paste
Salt :- Common salt, Baking soda.

b) Write two probable chemical tools (indicators) that provided by the lab demonstrator to the studies.

→ • Red litmus paper • Blue litmus paper

c) Observation is not sure test. Justify.

→ Observation may differ from person to person, so it cannot always give accurate results. Therefore, observation is not a sure test.

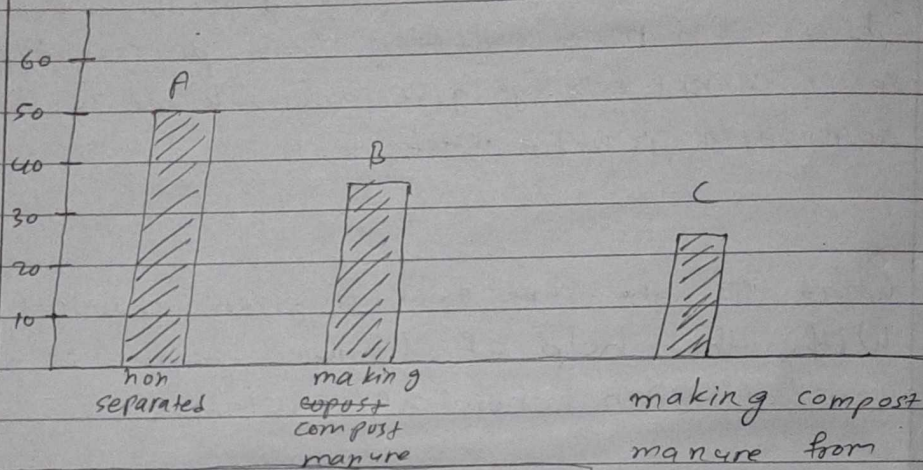
d) What is the sure test of given chemical substance?

→ With the help of litmus paper is the sure test of given chemical substances.

7. Lemon juice, soap water and common salt solution are tested with indicators like red litmus paper and blue litmus paper. They change indicators in different colours and chemical substances are identified. Study the above paragraph and fill the blanks given below.

Lemon juice shows
Red litmus with ~~blue~~ ^{lemon juice shows} ~~litmus~~ paper No
Colour and blue litmus shows red colour. Red
litmus with soap water shows blue colour and blue
litmus shows No colour. Red litmus with salt solution
water shows No colour and blue litmus
show blue colour. Hence lemon juice is an acid
soap water is a base and salt solution is salt.

8. The bar diagram of a survey given below is the analysis of data of solid waste management in Manara Sirwa municipality Mahottari. Study the bar diagram and answer the following questions.



- a) Mention the title of survey represent in the bar diagram.
→ To study the ~~solid waste~~ waste management in Manara Sirwa municipality.

degradable waste and other things from non-degradable waste.

- b) How many people send solid waste without separating in metropolitan vehicles?
→ 50 people send solid waste without separating in metropolitan vehicles.

- c) What represents the bar diagram 'C'?
→ The bar diagram 'C' represent the people who making copost manure from degradable waste and other things from non-degradable waste.

d) What is the conclusion of given bar diagram of the survey?

→ From the result it is found that people use recycle and reuse methods to manage the solid wastes in Mahara Siva municipality, Mahottari.

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* Unit - 1 *

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